WiFi Operation Instructions

■ File System Creation

The files required for WiFi operation have already been incorporated into in CD:¥rootfs¥rootfs-xscale-060924-2.tar.bz2.

They were incorporated using the following procedure.

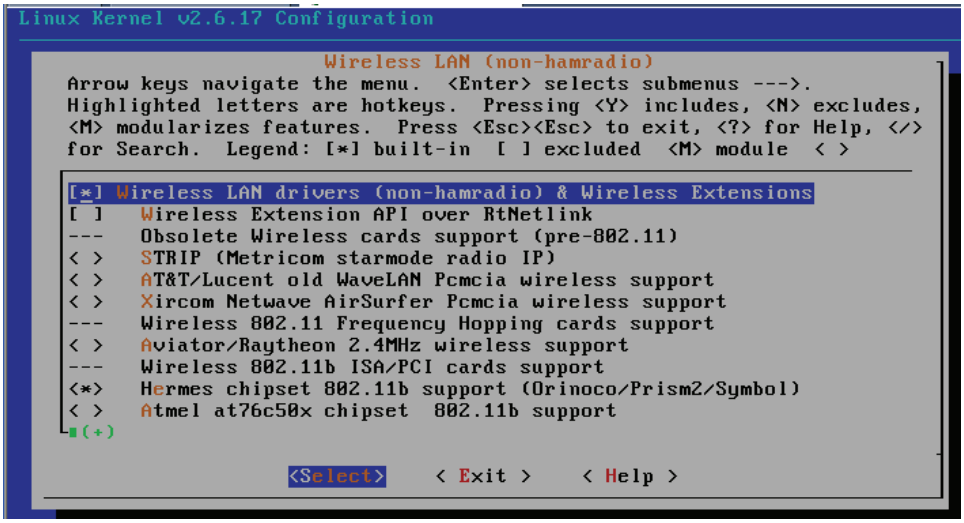
1. Copy the modules folder under rootfs-xscale-060924-2/lib.
3. Copy config and config.opt under rootfs-xscale-060924-2/etc/pcmcia.
4. Copy cardmgr and cardctl under rootfs-xscale-060924-2/sbin.

■ Kernel Construction

1. Follow the procedure in “5.2 Kernel Construction Procedure” in the Linux Installation Handbook for SandgateVP.pdf and execute up to “make menuconfig of 3.
2. Menu Configuration Check

Check if the Wireless LAN setting is as follows.

Device Drivers->Network device support->Wireless LAN(non-hamradio)



Select “Yes.”



3. Execute the “make” command.

■ WiFi Operation Procedure

1. Write the created kernel and file system to the Flash.
2. Use the following procedure to execute the commands in the serial console. The manually input commands are shown in red.

HyperTerminal

```
RedBoot(tm) bootstrap and debug environment [ROM]
Non-certified release, version Tiger.v004-004 - built 17:56:03, Sep 28 2006

Platform: PXA270 Sandgate-VP (XScale)
Copyright (C) 2000, 2001, 2002, Red Hat, Inc.

RAM: 0xa0000000-0xa4000000, 0xa000e1a8-0xa3f9d000 available
FLASH: 0x10000000 - 0x18000000, 512 blocks of 0x00040000 bytes each.
== Executing boot script in 3.000 seconds - enter ^C to abort
RedBoot> fis load zImage
RedBoot> exec -c "console=ttyS0,115200 mem=64M@0xa0000000 root=/dev/mtdblock1 ro
otfstype=jffs2"
Using base address 0xa1000000 and length 0x00200000
Uncompressing Linux.....................................................................
..... done, booting the kernel.
Linux version 2.6.17 (root@PXALinuxFC5) (gcc version 4.1.0) #1 PREEMPT Tue Oct 1
7 19:32:18 KST 2006
CPU: XScale-PXA270 [69054117] revision 7 (ARMv5TE)
Machine: Sophia Systems Sandgate-VP Development Platform
Memory policy: ECC disabled, Data cache writeback
Run Mode clock: 208.00MHz (*16)
Turbo Mode clock: 416.00MHz (*2.0, active)
Memory clock: 208.00MHz (/2)
System bus clock: 208.00MHz
CPU0: D VIVT undefined 5 cache
CPU0: I cache: 32768 bytes, associativity 32, 32 byte lines, 32 sets
CPU0: D cache: 32768 bytes, associativity 32, 32 byte lines, 32 sets
Built 1 zonelists
Kernel command line: console=ttyS0,115200 mem=64M@0xa0000000 root=/dev/mtdblock1
  rootfstype=jffs2
PID hash table entries: 512 (order: 9, 2048 bytes)
Console: colour dummy device 80x30
Dentry cache hash table entries: 8192 (order: 3, 32768 bytes)
Inode-cache hash table entries: 4096 (order: 2, 16384 bytes)
Memory: 64MB = 64MB total
Memory: 61884KB available (2417K code, 465K data, 96K init)
Mount-cache hash table entries: 512
CPU: Testing write buffer coherency: ok
NET: Registered protocol family 16
sandgate_vp_keypad_read_matrix: Data invalid(000000ff)
[sandgate_vp_i2c_read:875]
[sandgate_vp_i2c_read:875]
voltage: 0
NET: Registered protocol family 23
NET: Registered protocol family 2
IP route cache hash table entries: 512 (order: -1, 2048 bytes)
TCP established hash table entries: 2048 (order: 1, 8192 bytes)
TCP bind hash table entries: 1024 (order: 0, 4096 bytes)
```

```
TCP: Hash tables configured (established 2048 bind 1024)
TCP reno registered
PXA27x Dynamic Power Management
PXA27x DPM: Initial frequency is 416000kHz.
NetWinder Floating Point Emulator V0.97 (double precision)
JFFS2 version 2.2. (NAND) (C) 2001-2003 Red Hat, Inc.
Initializing Cryptographic API
io scheduler noop registered (default)
Console: switching to colour frame buffer device 30x40
SA1100 Real Time Clock driver v1.00
SA1100/PXA2xx Watchdog Timer: timer margin 60 sec
pxa2xx-uart.0: ttyS0 at MMIO 0x40200000 (irq = 21) is a BTUART
pxa2xx-uart.1: ttyS1 at MMIO 0x40100000 (irq = 22) is a FFUART
pxa2xx-uart.2: ttyS2 at MMIO 0x40700000 (irq = 20) is a STUART
loop: loaded (max 8 devices)
Uniform Multi-Platform E-IDE driver Revision: 7.00alpha2
ide: Assuming 50MHz system bus speed for PIO modes; override with idebus=xx
ide-floppy driver 0.99.newide
Probing Intel PXA27x On-Chip ROM at physical address 0x00000000 (32-bit bankwidt
h)
Intel PXA27x On-Chip ROM: Found 2 x16 devices at 0x0 in 32-bit bank
Intel PXA27x On-Chip ROM: Found 2 x16 devices at 0x4000000 in 32-bit bank
  Intel/Sharp Extended Query Table at 0x010A
  Intel/Sharp Extended Query Table at 0x010A
  Intel/Sharp Extended Query Table at 0x010A
  Intel/Sharp Extended Query Table at 0x010A
  Intel/Sharp Extended Query Table at 0x010A
Using buffer write method
cfi_cmdset_0001: Erase suspend on write enabled
Searching for RedBoot partition table in Intel PXA27x On-Chip ROM at offset 0x7f
c0000
5 RedBoot partitions found on MTD device Intel PXA27x On-Chip ROM
Creating 5 MTD partitions on "Intel PXA27x On-Chip ROM":
0x00000000-0x00040000 : "RedBoot"
0x00040000-0x03240000 : "jffs2"
0x03240000-0x03440000 : "zImage"
0x07f80000-0x07f81000 : "RedBoot config"
mtd: partition "RedBoot config" doesn't end on an erase block -- force read-only
0x07fc0000-0x08000000 : "FIS directory"
pxa27x_udc: version 23-Sep-2004
input: pxa27x-keypad as /class/input/input0
input: sandgatevp keypad driver!
input: pxa27x-pointer as /class/input/input1
input: sandgatevp pointer driver!
i2c /dev entries driver
enable_irq(18) unbalanced from c0018180
I2C: Adding PXA i2c adapter.
I2C: Scanning bus (02)(03)<5>pccard: PCMCIA card inserted into slot 1
pcmcia: registering new device pcmcia1.0
(04)(05)(06)(07)(08)(09)(0a)(0b)(0c)(0d)(0e)(0f)(10)(11)(12)(13)(14)(15)(16)(17)
(18)(19)(1a)(1b)(1c)(1d)(1e)(1f)(20)(21)(22)(23)(24)(25)(26)(27)(28)(29)(2a)(2b)
(2c)(2d)(2e)(2f)(30)(31)(32)(33)(34)(35)(36)(37)(38)(39)(3a)(3b)(3c)(3d)(3e)(3f)
(40)(41)(42)(43)(44)(45)(46)(47)(48)(49)(4a)(4b)(4c)(4d)(4e)(4f)(50)(51)(52)(53)
(54)(55)(56)(57)(58)(59)(5a)(5b)(5c)(5d)(5e)(5f)(60)(61)(62)(63)(64)(65)(66)(67)
(68)(69)(6a)(6b)(6c)(6d)(6e)(6f)(70)(71)(72)(73)(74)(75)(76)(77)(78)(79)(7a)(7b)
(7c)(7d)(7e)(7f)
I2C: Successfully added bus
I2C: Adding PXA power i2c adapter.
I2C: Scanning bus (14)(3b)(50)(51)(52)(53)
```

```
PWRI2C: Successfully added bus
I2C: PXA algorithm module loaded.
ac97_codec: AC97 Audio codec, id: AKM16 (Asahi Kasei AK4650)
TCP bic registered
NET: Registered protocol family 17
VFS: Mounted root (jffs2 filesystem).
Freeing init memory: 96K
INIT: version 2.86 booting
[STATE: 0] /etc/lids/lids.conf
[STATE: 1] /etc/lids/lids.boot.conf
        ACL Discovery is OFF
        Compiling into /etc/lids/lids.boot.acl
        Total 14 ACLs

[STATE: 2] /etc/lids/lids.postboot.conf
        ACL Discovery is OFF
        Compiling into /etc/lids/lids.postboot.acl
        Total 13 ACLs

[STATE: 3] /etc/lids/lids.shutdown.conf
        ACL Discovery is OFF
        Compiling into /etc/lids/lids.shutdown.acl
        Total 12 ACLs

INIT: Entering runlevel: 3

(none) login: root
login[769]: root login    on `ttyS0'

-bash-3.1#   insmod    /lib/modules/2.6.17/pcmcia/cfio.ko
cfio: module license 'unspecified' taints kernel.
-bash-3.1#   insmod    /lib/modules/2.6.17/pcmcia/mcf25.ko
Before register driver
After register driver
-bash-3.1#   cardmgr    -v
pcmcia: Detected deprecated PCMCIA ioctl usage from process: cardmgr.
pcmcia: This interface will soon be removed from the kernel; please expect break
age unless you upgrade to new tools.
pcmcia: see http://www.kernel.org/pub/linux/utils/kernel/pcmcia/pcmcia.html for
details.
cardmgr[799]: watching 2 sockets
cardmgr[800]: starting, version is 3.2.8
cardmgr[800]: initializing socket 1
cardmgr[800]: socket 1: Marvell 88W8300 802.11g PC Card
-bash-3.1# cardmgr[800]:     product info: "Murata Mfg.Co.,Ltd.", "RF-LEG2000Enter
ing cf_attach()
  W-LAN"
cardmgr[800]:     manfid: 0x02df, 0x8103    function: 6 (nLeaving cf_attach()
etwork)
BasePort1=0xcc8a0000, AssignedIRQ=46
After calling wlan_add_card function
cardmgr[800]: executing: './network start eth0'
cardmgr[800]: + sh: ./network: No such file or directory
cardmgr[800]: start cmd exited with status 127

-bash-3.1#   iwpriv    eth0    powercfg    0
eth0          powercfg:0    0    0    0
-bash-3.1#   iwconfig    eth0    txpower    11
-bash-3.1#   iwlist    eth0    scan
```

```
eth0        Scan completed :
            Cell 01 - Address: 00:03:2F:25:1A:41
                      ESSID:"QOSDEMO"
                      Mode:Managed
                      Frequency:2.437 GHz (Channel 6)
                      Quality:0/10    Signal level=-35 dBm    Noise level=-96 dBm
                      Encryption key:off
                      Bit Rates:54 Mb/s

-bash-3.1#  iwconfig  eth0  essid  QOSDEMO
-bash-3.1#  ifconfig  eth0  192.168.3.40  up
-bash-3.1#  ping  192.168.3.1
PING 192.168.3.1 (192.168.3.1): 56 data bytes
84 bytes from 192.168.3.1: icmp_seq=0 ttl=64 time=27.6 ms
84 bytes from 192.168.3.1: icmp_seq=1 ttl=64 time=9.9 ms
84 bytes from 192.168.3.1: icmp_seq=2 ttl=64 time=10.0 ms
84 bytes from 192.168.3.1: icmp_seq=3 ttl=64 time=10.0 ms
84 bytes from 192.168.3.1: icmp_seq=4 ttl=64 time=10.0 ms
84 bytes from 192.168.3.1: icmp_seq=5 ttl=64 time=10.0 ms
84 bytes from 192.168.3.1: icmp_seq=6 ttl=64 time=9.9 ms
84 bytes from 192.168.3.1: icmp_seq=7 ttl=64 time=10.0 ms
```